

Exam 2 (November 9/13)

QTM 220

Problem 1

Suppose we have a sample of $Y_1 \dots Y_n$ and $X_1 \dots X_n$ stored in an **R** dataframe 'df' with n rows. X can take the values of 1,2,3,4 or 5. We run this code.

```
B=10000
theta<- rep(NA,B)
for( b in 1:B ){
  bootind = sample(1:n,size=n,replace=T)
  dfboot = df[bootind,]
  theta[b] = (1/4)*mean(dfboot$y[dfboot$x==5]) - (1/4)*mean(dfboot$y[dfboot$x==1])
}
```

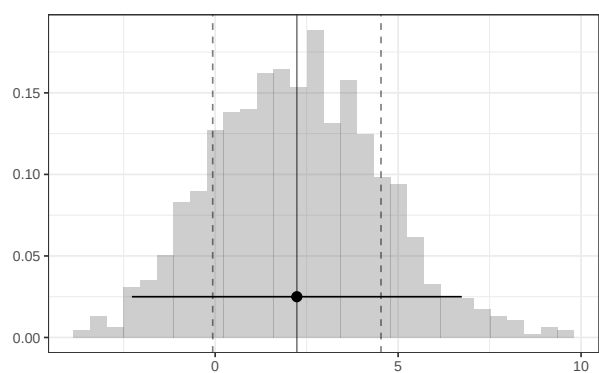


Figure 1: Histogram of the vector Y's elements

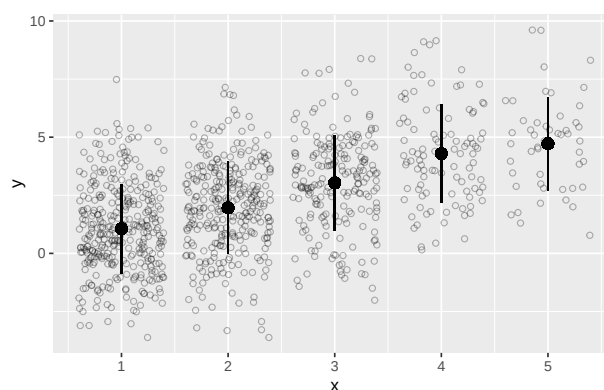


Figure 2: Jittered Scatterplot of X and Y

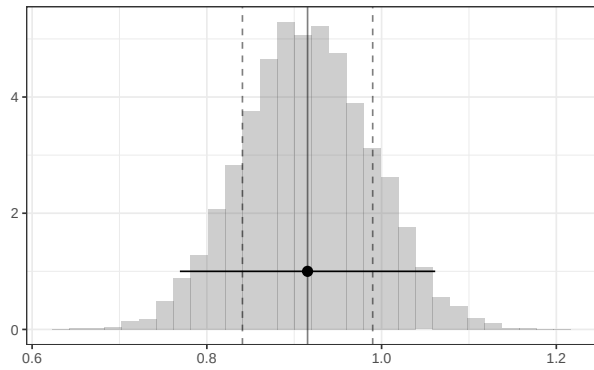


Figure 3: Histogram of the theta's elements

Part a

What does this code calculate? State it in words and mark it with an 'a', and an arrow or bracket if necessary for clarity, on the appropriate plot above.

```
mean(Y)
```

The sample mean (unconditioned). It's indicated by the solid vertical line on the sample histogram.

Part b

What does this code calculate? State it in words and mark it with a 'b', and an arrow or bracket if necessary for clarity, on the appropriate plot above.

```
sd(Y)
```

The sample standard deviation. It's the distance between the solid vertical and either dashed vertical line on the sample histogram.

Part c

What does this code calculate? State it in words and mark it with a ‘c’, and an arrow or bracket if necessary for clarity, on the appropriate plot above.

```
mean(Y[X==1])
```

The conditional sample mean of Y among X=1. It's indicated by the black dot on the scatterplot where X=1.

Part d

What does this code calculate? State it in words and mark it with a ‘d’, and an arrow or bracket if necessary for clarity, on the appropriate plot above.

```
sd(theta)
```

The bootstrap sampling distribution's standard deviation for the average increment estimator. It's the distance between the solid vertical and either dashed vertical line on the sampling distribution histogram.

Part e

What does this code calculate? State it in words and mark it with a ‘e’, and an arrow or bracket if necessary for clarity, on the appropriate plot above.

```
(1/4)(mean(Y[X=5]) - mean(Y[X=1]))  
- 1.96* sqrt((1/4)^2*var(Y[X=5])/length(Y[X=5]) + (-1/4)^2*var(Y[X=1])/length(Y[X=1]))  
(1/4)(mean(Y[X=5]) - mean(Y[X=1]))  
+ 1.96* sqrt((1/4)^2*var(Y[X=5])/length(Y[X=5]) + (-1/4)^2*var(Y[X=1])/length(Y[X=1]))
```

A 95% confidence interval — based on normal approximation — for the average incremental difference in Y from X=1 to X=5 in the population from which our data was drawn. This is indicated by

the horizontal segment (with a dot in the middle) on the sampling distribution histogram (theta).

Part f

How could you calculate roughly the same thing as in part e using 'sd(theta)' and not

```
sqrt(((1/4)^2*var(Y[X=5])/length(Y[X=5])) + ((-1/4)^2*var(Y[X=1])/length(Y[X=1])))
```

We can use the standard deviation of the bootstrap sampling distribution as our estimate of standard error.

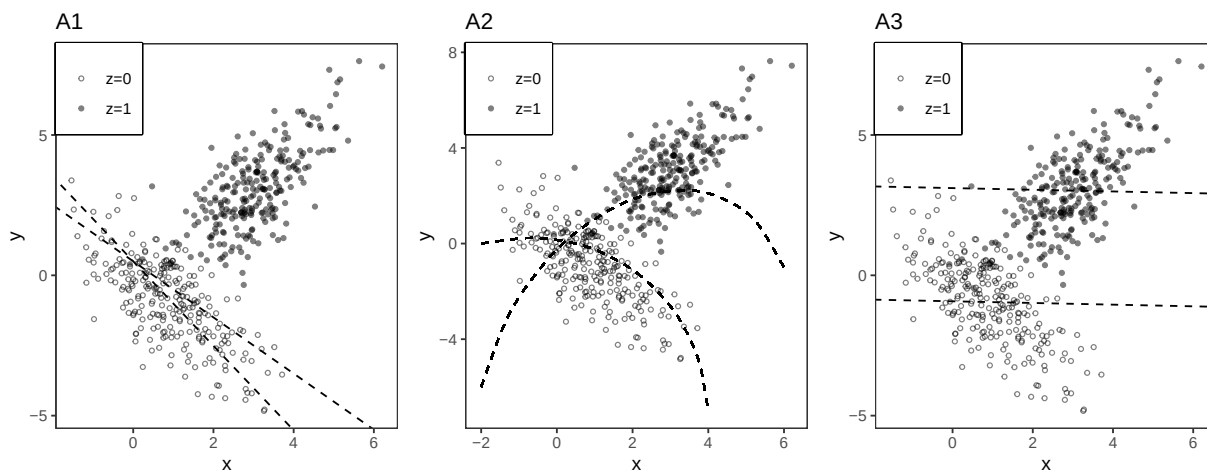
```
mean(Y) - 1.96 * sd(theta)
mean(Y) + 1.96 * sd(theta)
```

Problem 2

In the following parts, choose the figure with the appropriate regression line(s).

Part A

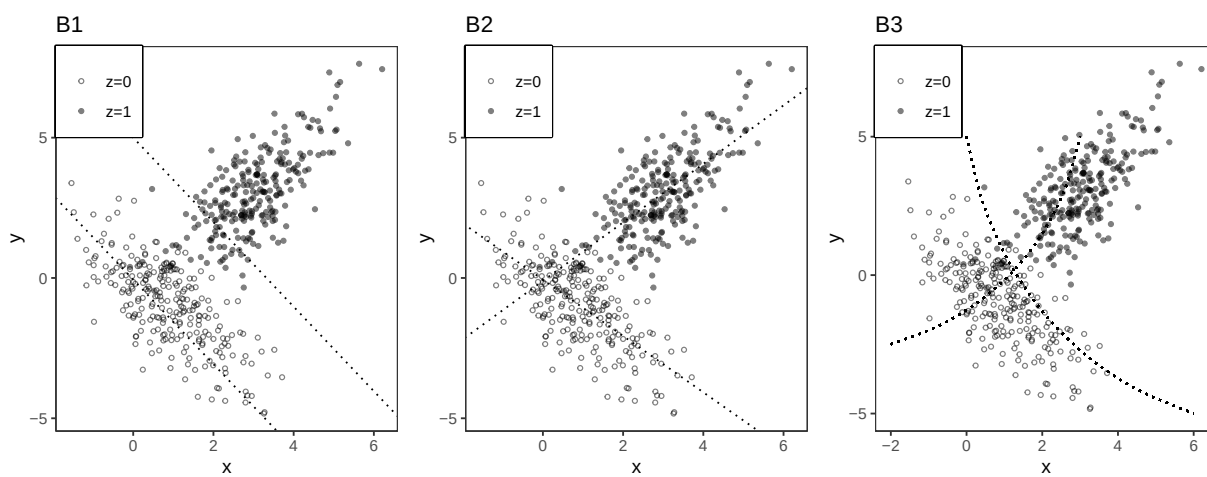
Choose the figure with the two dashed lines associated with the additive multiple regression ($y \sim x + z$).



A3

Part B

Choose the figure with the two dotted lines associated with the interactive multiple regression ($y \sim x * z$).



B2

Problem 3

Suppose we want β_1 from $y_i = \beta_0 + \beta_1 x_i + \beta_2 z_i + \beta_3 w_i + e_i$ but we run the regression $lm(y \sim x)$ without z or w . We have de-meaned our variables x_i , z_i , and w_i such that $\bar{x} = 0$, $\bar{y} = 0$, and $\bar{w} = 0$. Under what condition(s) will the coefficient on x from $lm(y \sim x)$ be equal to β_1 ? Derive what your coefficient on x will be in terms of β s and justify your steps.

The coefficient on x from $lm(y \sim x)$ is

$$\frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2}$$

Plug in y_i defined by the multivariate regression: $y_i = \beta_0 + \beta_1 x_i + \beta_2 z_i + \beta_3 w_i + e_i$

$$\frac{\sum_{i=1}^n x_i (\beta_0 + \beta_1 x_i + \beta_2 z_i + \beta_3 w_i + e_i)}{\sum_{i=1}^n x_i^2}$$

Distribute

$$\frac{\sum_{i=1}^n x_i \beta_0}{\sum_{i=1}^n x_i^2} + \frac{\sum_{i=1}^n x_i \beta_1 x_i}{\sum_{i=1}^n x_i^2} + \frac{\sum_{i=1}^n x_i \beta_2 z_i}{\sum_{i=1}^n x_i^2} + \frac{\sum_{i=1}^n x_i \beta_3 w_i}{\sum_{i=1}^n x_i^2} + \frac{\sum_{i=1}^n x_i e_i}{\sum_{i=1}^n x_i^2}$$

Let's work through each term.

$$\frac{\sum_{i=1}^n x_i \beta_0}{\sum_{i=1}^n x_i^2} = \frac{\beta_0 \sum_{i=1}^n x_i}{\sum_{i=1}^n x_i^2} = \frac{\beta_0 * 0}{\sum_{i=1}^n x_i^2} = 0$$

as x is demeaned, $\sum x_i = 0$

$$\frac{\sum_{i=1}^n x_i \beta_1 x_i}{\sum_{i=1}^n x_i^2} = \frac{\beta_1 \sum_{i=1}^n x_i x_i}{\sum_{i=1}^n x_i^2} = \beta_1$$

$$\frac{\sum_{i=1}^n x_i \beta_2 z_i}{\sum_{i=1}^n x_i^2} = \frac{\beta_2 \sum_{i=1}^n x_i z_i}{\sum_{i=1}^n x_i^2}$$

$$\frac{\sum_{i=1}^n x_i \beta_3 w_i}{\sum_{i=1}^n x_i^2} = \frac{\beta_3 \sum_{i=1}^n x_i w_i}{\sum_{i=1}^n x_i^2}$$

$$\frac{\sum_{i=1}^n x_i e_i}{\sum_{i=1}^n x_i^2} = \frac{0}{\sum_{i=1}^n x_i^2} = 0$$

we know that $\sum_{i=1}^n x_i e_i = 0$ from the first order conditions on our least squares estimator (we chose the intercept and coefficient(s) that minimize the sum of squared residuals)

Thus we have

$$\frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2} = \beta_1 + \frac{\beta_2 \sum_{i=1}^n x_i z_i}{\sum_{i=1}^n x_i^2} + \frac{\beta_3 \sum_{i=1}^n x_i w_i}{\sum_{i=1}^n x_i^2}$$

The coefficient on x from the simple regression of y on x will be equal to β_1 if: (1) β_2 or the coefficient on x from the simple regression from z on x is 0 and (2) β_3 or the coefficient on x from the simple regression of w on x is 0. If either of these terms is non-zero, our simple regression will not give us the same result as our regression that includes both x , z , and w . If x is randomized (like a random treatment assignment), it should not be correlated with z or w , so our short regression coefficient should give us β_1 .

Problem ...

FWL problem?

unbiasedness proof?

causal (balance? multi-level treatment? coefficient on treatment as ATE?)